CIE 9702 Physics Paper 4 Style Question

Topic: Gravitation and Orbital Motion (Neptune System)

46 The moons of Neptune

The space probe Voyager 2 flew through the Neptunian system on 24 August 1989. It had been travelling for 12 years, covered more than 7 billion kilometres and passed Jupiter, Saturn and Uranus. Most of the energy needed was given to the space probe at its launch, to get it away from the Earth. It then used the gravitational pulls of each planet in turn to help it get further and further away from the sun.

It was already known that Neptune had two moons, Triton and Nereid, but Voyager 2 discovered six more. Details of them are given in the table. Note that 1989N1 is *larger* than Nereid.

moon	diameter/km	period/hr	orbital radius/10^3 km
new:			
1989N1	420	26.9	117.6
1989N2	200	13.3	73.6
1989N3	140	8.0	52.5
1989N4	160	9.5	60.0
1989N5	90	7.5	50.0
1989N6	50	7.1	48.2
previously known:			
Triton	2720	140.9	354.3
Nereid	340	8640	5500

All the newly discovered moons are very cratered and irregular in outline. Only the largest one, 1989N1, is close to the size above which its outline would be rounded off by its own gravity.

In the following questions you may need to know that the Universal Gravitational Constant, $G = 6.67 \times 10^{-11}\,\mathrm{Nm^2kg^{-2}}$, and that the force F between two masses m_1 and m_2, distance r apart is given by $F = Gm_1m_2/r^2$.

(a) List the changes in potential and kinetic energy that a space probe goes through from launch to 'parking orbit', then as it moves away from Earth, and finally as it flies past a planet. *(4)*

(b) Sketch the path taken by a space probe as it flies past a planet, and indicate on it where the kinetic energy is greatest. If no fuel is used where does this energy come from? *(4)*

(c) Express the Earth's gravitational field strength g in terms of G, the mass of the Earth M and the distance from the centre of the Earth r. *(2)*

(d) Given that $g = 9.81\,\mathrm{Nkg^{-1}}$ at the surface of the Earth, whose radius is 6370 km, calculate the mass of the Earth. *(2)*

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(e) Calculate the energy which must be supplied to each kilogram of a space probe to enable it to escape totally from the Earth's pull. *(3)*

(f) By using the other information in the table, give a reason why 1989N1 was not discovered from Earth, even though it is bigger than Nereid. *(2)*

(g) By considering the gravitational forces on projections on an irregular but very large moon, suggest why such a moon would be 'rounded off'. *(3)*

(h) Assume that the orbits of Neptune's moons are circular. By equating the expression for the centripetal force on a moon with the gravitational attraction towards Neptune, show that the radius r and the period T of the orbit are related by

$$\frac{r^3}{T^2} = \frac{GM_N}{4\pi^2}$$

where M_N is the mass of Neptune. *(4)*

(i) Use the data in the table for 1989N1 and for Nereid with the expression in (h) to calculate the mass of Neptune. Suggest why the answers you get are not exactly the same, though the expression in (h) implies that they are. *(6)*

Total 30 marks

85

Passage

In August 1989, the Voyager 2 space probe entered the Neptunian system after a journey of approximately 12 years. Mission planning required precise use of gravitational fields to alter the probe's speed and direction while minimising fuel consumption. During its fly-by, Voyager 2 identified several new moons orbiting Neptune, in addition to the previously known moons Triton and Nereid.

Scientists were interested in how orbital period T depends on orbital radius r for moons of the same planet. If Neptune has mass M_N, then combining uniform circular motion with Newtonian gravitation gives

$$\frac{r^3}{T^2} = \frac{GM_N}{4\pi^2}$$

where $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$.

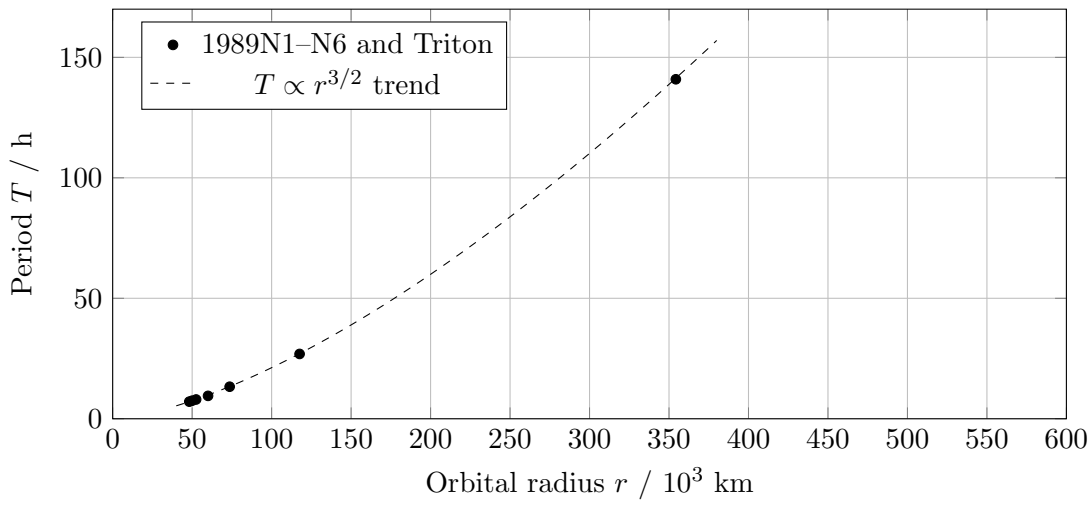
Table 1 gives measured values for six newly observed moons and two previously known moons.

Table 1: Moons of Neptune

Moon	Diameter / km	Period, T / h	Orbital radius, r / 10^3 km	T^2/r^3 / 10^{-8} h^2 km^{-3}
1989N1	420	26.9	117.6	1.12
1989N2	200	13.3	73.6	1.17
1989N3	140	8.0	52.5	1.11
1989N4	160	9.5	60.0	1.11
1989N5	90	7.5	50.0	1.13
1989N6	50	7.1	48.2	1.13
Triton	2720	140.9	354.3	1.12
Nereid	340	8640	5500	4.95

For one of the newly discovered moons, the probe measured an orbital speed of about $1.95\times10^3\,\mathrm{m\,s^{-1}}$ at radius $r = 5.25 \times 10^7$ m.

Figure 1: Orbital period against orbital radius for selected moons



Questions

(a) Describe the changes in gravitational potential energy and kinetic energy of a space probe:

- from rest on the Earth to a low circular parking orbit,
- from parking orbit to deep space during escape,
- during a planetary fly-by used as a gravity assist.

[4]

(b) By considering force balance for circular motion, derive

$$\frac{r^3}{T^2} = \frac{GM_N}{4\pi^2}$$

for a moon of mass m orbiting Neptune. [4]

(c) Using moon 1989N3 data ($T = 8.0\,\mathrm{h}$, $r = 52.5 \times 10^3\,\mathrm{km}$), calculate the mass of Neptune. [3]

(d) Repeat the calculation in (c) using Triton data. Comment on one reason the two values may differ slightly. [3]

(e) Use the additional speed measurement $v = 1.95 \times 10^3\,\mathrm{m\,s^{-1}}$ at $r = 5.25 \times 10^7\,\mathrm{m}$ to obtain a third estimate of Neptune's mass from $v^2 = GM_N/r$. Compare with your answers in (c) and (d). [3]

(f) Figure 1 suggests a power-law relation between T and r.

- Explain why plotting $\log T$ against $\log r$ would help test this model.
- State the expected gradient and intercept form in terms of constants.

[4]

(g) Nereid does not lie close to the trend followed by most other points. Suggest, with reference to orbital shape and perturbations, why this moon may behave differently. [3]

(h) *Open-ended:* A future probe is planned to orbit Neptune and then descend to study Triton. Propose one mission profile that reduces fuel usage while maintaining mission safety. Your answer should include at least two physics principles from this question and one practical engineering consideration. [6]

Suggested answers (teacher support)

(a) GPE increases (becomes less negative) as r increases; KE rises to achieve orbital speed; for escape, KE and total mechanical energy increase to zero/positive total energy; gravity assist can increase heliocentric KE by exchanging energy-momentum with a moving planet.

(b)

$$\frac{GM_N m}{r^2} = m\frac{v^2}{r}, \quad v = \frac{2\pi r}{T}$$
$$\Rightarrow \frac{GM_N}{r^2} = \frac{4\pi^2 r}{T^2} \Rightarrow \frac{r^3}{T^2} = \frac{GM_N}{4\pi^2}$$

(c) $T = 2.88 \times 10^4\,\mathrm{s},\ r = 5.25 \times 10^7\,\mathrm{m}$

$$M_N = \frac{4\pi^2 r^3}{GT^2} \approx 1.04 \times 10^{26}\,\mathrm{kg}$$

(d) $T = 5.0724 \times 10^5\,\mathrm{s},\ r = 3.543 \times 10^8\,\mathrm{m}$

$$M_N \approx 1.02 \times 10^{26}\,\mathrm{kg}$$

Difference may arise from rounding, measurement uncertainty, non-circularity, or perturbations by other moons.

(e)

$$M_N = \frac{v^2 r}{G} = \frac{(1.95 \times 10^3)^2 (5.25 \times 10^7)}{6.67 \times 10^{-11}} \approx 3.0 \times 10^{24}\,\mathrm{kg}$$

This is far lower than (c) and (d), suggesting the speed value is likely local/instantaneous on a non-circular path or not matched to the same moon-state assumptions.

(f) If $T = kr^n$, then $\log T = n \log r + \log k$. A log-log graph gives a straight line if the power law is valid. Expected gradient $n = 3/2$ and intercept $\log k$.

(g) Nereid has a strongly eccentric and inclined orbit; average radius is a poorer description of its motion; stronger perturbative effects can shift observed period-radius relation from simple circular model.

(h) Any justified proposal: e.g. gravity assists + staged insertion burns + high-periapsis capture then repeated moon fly-bys to lower apoapsis; include orbital-energy arguments, Δv minimisation, radiation/communication/thermal constraints.